Field-Theoretic Criterion for DRAM High-k Dielectrics: A Two-Dimensional Criterion Space (Soft-Mode × Bandgap) and Quantitative Doping-Pinning

Zenodo preprint v2.1 (English) | 2026-08-18 Author: Chao Qin | ORCID 0009-0006-2000-5644 | Juexiao Information Consulting Center, Xingyi, Guizhou 562400, China **Category**: DRAM capacitor dielectrics / materials criterion / elastic soft modes / antiferroelectrics / data-driven materials screening / unified storage-medium criterion **Chinese version**: released simultaneously

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Abstract

DRAM capacitor dielectrics face a hard constraint of leakage×EOT scaling below the 1z-nm node (Shiratake, IMW 2020). For 30 years industry has responded by doping-pinning the tetragonal phase of high-k dielectrics (ZrO⊠/HfO⊠-based) to stabilize it—yet this long-standing practice lacks a unified criterion language. Using a field-theoretic framework (materials as density-cluster fields ρ=m·p; phase-transition candidates as failure-line R'=1 equilibrium regions; shear soft modes as signatures of phase-transition activity), this paper provides a computable criterion coordinate for DRAM high-k dielectrics. Core results: (1) **Industry hit validation of the soft-mode criterion**—in a 6-material comparison of the high-k family, ZrO⊠, the only material with antiferroelectric (AFE) transition activity, is exactly the only Hf/Zr oxide that falls in the soft-mode region (C44=20.3 GPa⩽threshold 22.7), while all other stable dielectrics (HfO⊠ 103.2/Al⊠O⊠ 43.8/SiO⊠ 67.1) are not soft (positive 1/1, negative 4/4); (2) **Two-dimensional criterion space**—soft-mode axis (transition activity / 0K metastability) × bandgap axis (leakage wall, threshold ~4 eV), with tetragonal ZrO⊠ P4⊠/nmc the unique Pareto-optimal phase (C44=30.7+Eg=4.03), explaining 30 years of industrial pinning of the tetragonal phase; (3) **Quantitative doping-pinning**—an alkaline-earth AZrO⊠ calibration family yields C44=83.7·r−30.1 GPa (R^2=0.9998), making dopant ionic radius a quantitative tuning knob of the C44 spectrum; Y³⊠ extrapolation 55.3 GPa falls within the measured 8YSZ range 47-66 GPa; (4) **Martensitic mirror**—the same t→m transformation is exploited in ceramics (toughening) but is a failure mode in DRAM dielectrics, so the criterion is invariant while engineering goals are opposite; (5) **Criterion D data engineering**—matching-rule auditing found 44.3% multi-entry contamination; an energy-window criterion (non-negative min-C44 within +0.1 eV of ground state) makes the criterion semantics explicit as "0K instability signal of a real metastable phase," with a global recomputation of a 546-compound list released. All values are machine-verified, with experimental comparisons carrying verifiable references.

Keywords: DRAM capacitor dielectric; high-k dielectric; shear soft mode; antiferroelectric; doping pinning; elastic criterion; materials screening

1. Introduction

Scaling of DRAM capacitor dielectrics is a long-standing consensus problem at the semiconductor physics limit (the "memory wall" material root hit by three independent industry-deficit lists; IMW 2020 explicitly states new materials are needed below the 1z-nm node [1]). EOT (equivalent oxide thickness) scaling hits the tunneling-leakage wall, and industry responds by high-k replacement—SiO⊠(k$_{\text{3.9)→Al⊠O⊠(k}}$9)→Ta⊠O⊠(k$_{\text{25)→ZrO⊠/HfO⊠(k}}$20-40)→ZAZ stack→next-generation TiO⊠(k>80)/antiferroelectric [2]. But high-k introduces two new problems:

- **Phase-transition activity**: ZrO⊠ AFE↔FE electric-field-induced switching enhances charge storage, but AFE/FE transitions bring polarization stability and fatigue issues [3];
- **Phase stability**: the high-k phases (tetragonal/cubic) of HfO⊠ are unstable at room temperature and must be stabilized by doping (an industry-standard method) [4].

Both are **phase-transition/phase-stability problems**—precisely the domain of a materials criterion layer. This paper' s position: industry' s 60-year doping-pinning practice is **position control on the C44 elastic spectrum**—making this tacit knowledge explicit through a computable criterion coordinate.

Framework (field theory): materials as density-cluster fields ρ=m·p (m=concentration, p=distribution shape); phase-transition candidates as failure-line R' =1 equilibrium regions (near-degenerate systems = criterion-amount protocol-sensitive regions); **shear soft mode C3** as the phonon-softening signature of displacive phase transitions (AFE↔FE switching activity). Using the SPTF data layer (V12/JARVIS elasticity), this paper maps DRAM high-k dielectrics onto these two criterion axes.

2. Methods and Data

2.1 Criterion quantities

- **Soft-mode axis (C3)**: minimum shear elastic constant C44 (min-C44)—small C44 = soft against shear = active displacive transition. Threshold C44⩽22.7 GPa (p25 percentile of the v0.9 positive sample).
- **Bandgap axis (leakage wall)**: DFT bandgap Eg—leakage current ∝exp(−Eg), with DRAM leakage specs requiring Eg⩾4 eV (industry empirical value).
- **Criterion D (data quality)**: matching-rule auditing found multi-entry JARVIS structures for the same compound (44.3% of 546 matched compounds are multi-entry), and the old criterion was contaminated by bad entries. Fixed criterion D = non-negative min-C44 within an energy window (ground-state formation energy + 0.1 eV)—making the criterion semantics explicit as "**0K instability signal of a real metastable phase**" (transition activity, not ground-state instability, nor hypothetical-phase noise).

2.2 Data sources

- JARVIS-DFT 3D elastic library (elastic tensor + formation energy + bandgap, local mirror)
- SPTF cross-channel joint ranking v1.0 (729 candidates, 546 with elasticity data)
- OQMD structure enumeration (ZrO⊠ polymorph 361 phases)
- Calibration family: alkaline-earth AZrO⊠ (same-stoichiometry perovskite, Shannon radius CN=8)

All values machine-verified (python3); scripts released in a reproducibility package.

3. Results

3.1 Industry hit validation of the soft-mode criterion

Six-material comparison of the high-k dielectric family (gun 2, corrected to criterion-D values):

Material	Industry role	C44(GPa)	Soft-mode	Industry behavior
ZrO⊠	Mainstay + AFE star	**20.3**	**hit**	has AFE activity √
HfO⊠	doping-stabilized mainstay	103.2	not soft	no AFE as pure phase, needs doping √
Al⊠O⊠	first-gen / ZAZ spacer	43.8	not soft	no transition activity (stable spacer) √

Material	Industry role	C44(GPa)	Soft-mode	Industry behavior
SiO⊠	old dielectric (discarded)	67.1 (quartz)	not soft	no activity + low k, discarded √
TiO⊠	next-gen candidate (k>80, unproduced)	34.3	not soft	bandgap wall (§3.3) √
Si⊠N⊠	old dielectric	100	not soft	stable √

> **Caliber note**: C44 in this table uses the industry-comparison caliber (c44_old, industry dielectric phase). Under criterion D (energy window, softest metastable phase) values differ: HfO⊠ tetragonal metastable 22.9 is near-soft (its "not soft as pure phase" is refined to "monoclinic ground state hard, tetragonal metastable near-soft"—the criterion basis for industry's heavy-doping need, gun 5); SiO⊠ metastable (P4⊠/mmc) 2.4 is an independent 0K-instability signal of a metastable phase (same class as the ZrO⊠ tetragonal metastable), not changing the "quartz SiO⊠ as a stable dielectric" industry comparison. Criterion-D soft-mode hit semantics = "0K instability signal of a real metastable phase" (§3.5).

Positive 1/1 hit, negative 4/4 correct—the only material with AFE activity is exactly the only soft-mode hit. Among the 56 soft-mode candidates (criterion D), only 2 are Hf/Zr oxides (HfLi⊠O⊠ artifact-flagged + ZrO⊠ the only real one)—**the soft-mode criterion uniquely selects ZrO⊠ from the Hf/Zr family**.

> Honest boundary: the n=6 comparison is an exhaustive industry-family check rather than a statistical sample; "hit rate" is a directional validation.

3.2 Two-dimensional criterion space: soft-mode × bandgap

DRAM high-k criterion space = soft-mode axis (transition activity) × bandgap axis (leakage wall):

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              bandgap axis (leakage wall, threshold ~4 eV)

Al2O3 6.43     high-wall region
HfO2 tetr 4.72 ← high-wall + soft (22.9) = better but more unstable
ZrO2 tetr 4.03 ←  Pareto-optimal: soft (30.7) + high bandgap
HfO2 mono 4.12 hard (95.6)
ZrO2 fluor 3.45  hard (65.5)
Ta2O5 3.1      edge
TiO2 rutile1.77  excluded (bandgap wall)
                        soft-mode axis →
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Key findings: 1. **Tetragonal ZrO⊠ P4⊠/nmc (JVASP-350, lowest-energy phase) has bandgap 4.03, the highest among ZrO⊠ phases**—the tetragonal phase is the Pareto-optimal point (soft-mode AFE activity + lowest leakage). This gives a complete criterion explanation of why industry has pinned the tetragonal phase for 30 years rather than using the more stable monoclinic phase: monoclinic is hard but has a lower bandgap and no AFE activity. 2. HfO⊠ tetragonal (C44=22.9/Eg=4.72) has a higher bandgap in 2D space but is softer at 0K—industry HfO⊠ needs heavier doping (differing from ZrO⊠'s "as-deposited tendency to tetragonal" [2]) = another stability-activity tradeoff dimension. 3. Bandgap threshold: Al⊠O⊠ 6.43/HfO⊠ 4.72/ZrO⊠ 4.03 pass, Ta⊠O⊠ 3.1 edge, TiO⊠ 1.77 excluded—corresponding one-to-one with the industry evolution history.

TiO⊠'s bandgap wall (gun-4 de-hallucination flip): the earlier reading "TiO⊠ too soft, hits wall (-16.5)" was a matching-entry artifact (P-62m hypothetical phase, Ef 0.25 eV higher); the rutile lowest-energy phase has C44=117.9 (hard). The real wall TiO_2 hits is the bandgap wall—rutile Eg=1.77 vs ZrO_2 4.03, a 3-4 order-of-magnitude higher leakage, which has kept TiO_2 out of DRAM for 20 years. Moreover TiNbO⊠ has zero bandgap (Nb doping metallization dead end).

> Honest boundary: bandgaps are JARVIS OptB88vdW values (DFT underestimation), and the absolute 4 eV threshold is an industry empirical value rather than a DFT value—the 2D axis values are DFT relative ordering; the threshold line awaits experimental calibration.

3.3 Quantitative doping-pinning: ionic-radius-C44 calibration curve

Alkaline-earth $AZrO_3$ calibration family (same-stoichiometry perovskite, joint-ranking C44 × Shannon radius CN=8):

Dopant ion	r(Å)	C44(GPa)
Mg^{2+}	0.89	44.3
Ca^{2+}	1.12	64.0
Sr^{2+}	1.26	75.1
Ba^{2+}	1.42	88.8

$C44 = 83.7 \cdot r - 30.1$ (R^2=0.9998)—dopant ionic radius maps almost perfectly linearly to C44. Doping is a quantitative tuning knob of the C44 spectrum.

Extrapolation and structure-domain boundary (tolerance factor $t=(r_A+r_O)/\sqrt{2}(r_B+r_O)$, r_O=1.40/r_B=0.84):

Ion	r(Å)	extrapolated C44	structure domain
Y^{3+}	1.02	55.3	t=0.76 boundary
Gd^{3+}	1.05	57.8	t=0.77 boundary
La^{3+}	1.16	67.0	t=0.81 distorted perovskite (extrapolation valid)

- **Soft-mode threshold r=0.63 Å lies below the tolerance-factor lower bound 0.77**—no dopant ion within the perovskite family can reach the soft mode: perovskite-domain doping can only pin (stiffen), not enhance activity (structural conclusion).
- **Y-doping dual structure domain**: fluorite solid-solution domain = stabilization (measured YSZ C44 47-66 GPa, extrapolation 55.3 hits [5]); perovskite domain = 0K instability ($YZrO_3$ C44=1.1 at tolerance boundary t=0.76)—industry YSZ uses the fluorite domain as a structural necessity.
- Extrapolation valid in the perovskite family (Mg/Ca/Sr/Ba/La); the fluorite solid-solution domain (Al/Si doping) is not extrapolable (different structure domain).

3.4 Martensitic mirror: invariant criterion, opposite engineering goals

The same t→m martensitic transformation (4% volume expansion + shear) [6]:

- **Ceramic toughening** = exploiting transition activity (ZrO_2 PSZ/TZP toughening ceramics; ZrO_2 is a ceramic star);
- **DRAM dielectrics** = failure (volume expansion + dielectric drop → leakage/EOT deterioration), with industry doping the t phase to pin it against transformation.

Same material, same criterion, mirrored use in two industries—C3 soft mode selects "transition activity," and the engineering use determines the value of that activity. This explains the dual identity of ZrO_2 as both a ceramic-toughening material and a DRAM high-k dielectric.

3.5 Criterion D: matching-rule audit and data engineering

- **Audit finding**: 44.3% (242/546) of matched compounds are multi-entry (multiple JARVIS structures per compound); the old criterion's soft-mode set was contaminated by bad entries—TiO⊠ once matched a P-62m hypothetical phase (triggered by gun 4).
- **Criterion D established**: energy window (within +0.1 eV of ground state) and non-negative min-C44, excluding both hypothetical-phase noise (energy window) and Born-instability bad values (non-negative filter).
- **Impact**: 31 of 546 compounds flipped; 24 marked as missing data; 55/56 resonance candidates preserved (only ZnTiO⊠ out)—the resonance set is highly robust.
- **Global recomputation** (gun 6): new 546-compound list released (c44_old/c44_d/d_jid/d_spg full fields); window-sensitivity scan (ΔE=0.05→139/0.10→152/0.20→163/0.30→164, monotonic convergence, 0.10 chosen as mid-range reasonable); 10 small-positive anomalies (c44_d<1 GPa) handled: 5 high-confidence bad values removed (diamond C44=0.1 vs true ~500—hard evidence/ spinel/BeF⊠/ice/Sc⊠O⊠) + 5 audit-candidate flags (release-blocking item ①, executed 8-17).
- **Criterion semantics final**: ZrO⊠'s soft-mode hit = 0K instability signal of the tetragonal polymorph family (family-level), not of a single industry phase (P4⊠/nmc itself C44=30.7, not soft)—the industry AFE transition is a specific phonon-mode softening (E-field-induced P4⊠/nmc→Pca2⊠ [3]), while global C44 shear-soft is a family-level activity proxy.

4. Discussion

Why industry has pinned the tetragonal phase for 30 years: the 2D criterion space gives the complete explanation—the tetragonal phase is the unique Pareto optimum (soft-mode activity + highest bandgap); industrial "doping stabilization" = pulling the working point back into the controllable region on the C44 spectrum (chemical pinning: quantified by the calibration curve; gradient pinning: ZAZ stack = interlayer C44-spectrum gradient [7]). DRAM high-k evolution = controllable progress on the soft-mode axis: SiO⊠(67.1 hard)→Al⊠O⊠(43.8)→ZrO⊠(20.3 soft + high k + AFE)→TiO⊠(bandgap wall)—each generation moves toward the soft end for k and activity, at the cost of stability (rising pinning demand).

Field-theoretic increment: this is not a new-material discovery but a criterion coordinate for industrial practice—isomorphic to the photoresist line's "monopoly = tacit tuning knowledge": industry's 60-year doping experience has a sortable coordinate on the C44 spectrum (ionic-radius calibration curve), providing a criterion starting point for next-generation dielectric screening (ferroelectric memory / PCM / 3D NAND charge-trap).

5. Unified Criterion for Storage Dielectrics (v2.0 extension)

The 2D criterion space for DRAM high-k dielectrics (soft-mode × bandgap) is part of a unified criterion for storage media. Extending the framework to ferroelectric memory (S-C2), phase-change memory (S-C3), and charge-trap memory (S-C4) yields the **five-dimensional unified storage-medium criterion**:

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Unified storage-medium criterion = Transition activity × Transition-activity signature
                                  × Transition limit × Transition retainability/post-drift
                                  × Retention quality

Dim 1 Transition activity (soft-mode C44):       S-C1 - how readily displacive transitions occur
Dim 2 Transition-activity signature (polar near-degeneracy): S-C2 - the specific FE/AFE channel
Dim 3 Transition limit (T_m = force-ratio critical): S-C3 - the "strongest event" (melting/boundary diss
Dim 4 Retainability/post-drift (T_x/ ):          S-C3 - stability and drift of the post-transition state
Dim 5 Retention quality (A=1/ ):                 S-C4 - adsorption strength (reciprocal dwell time)
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Shared "force-ratio critical / retention" language (the basis of unification across the four storages): - **S-C1**: failure line R'=1 (near-degenerate systems = criterion-protocol-sensitive region) - **S-C2**: polar channel t↔o near-degeneracy (~1 meV/f.u.) — the ferroelectric phase Pca2⊠ itself is not soft (HfO⊠ C44=94.0/ZrO⊠ 82.8); ferroelectric activity = soft-mode precursor (tetragonal soft, HfO⊠ 22.9/ZrO⊠ 20.3) + polar-channel near-degeneracy (HfO⊠ Pca2⊠ ΔE=+0.0177=FE / ZrO⊠ t↔o 1 meV=AFE) - **S-C3**: T_m = force-ratio critical temperature (Lindemann≡Born equivalence) — the static screening quantity $L=\Theta\_D^2 \cdot M \cdot V$^(2/3) completely orders GST pseudo-line T_m (Spearman ρ=1.000) - **S-C4**: adsorption strength A=1/τ (reciprocal dwell time) — the CTF retention model (Arrhenius with an activation energy embedded in τ) is itself an engineering realization of A=1/τ; experiments confirm A=1/τ is a distribution (widely distributed time constants)

★ **Retention spectrum (v2.1 core extension)**: the unified criterion extends from a "transition spectrum" to a "retention spectrum"—phase-change memory and charge-trap memory share the same adsorption-strength coordinate A=1/τ:

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Decay-shape dichotomy (v0.4 §4):
  Exponential end (CTF, stretched exponential): ΔVth ~ exp[-(t/ )^ ] - charge escape, Arrhenius activatio
  Power-law end (PCM):                           ΔR ~ (t/t )^ ,  0.05-0.12 - structural relaxation, FDNE s
→ The two media are discriminable by the time-dependence exponent (exponential: semi-log linear
  vs power-law: log-log linear) - two realizations of the same A=1/  coordinate, two ends of retention q
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Four storages = **different segments of the same retention spectrum**:

Storage	Retention mode	Criterion coordinate	Application
DRAM high-k (AFE charge)	reversible transition in the near-degenerate region	soft mode 20.3 + bandgap 4.03 (ZrO⊠)	charge storage
Ferroelectric memory (FE polarization)	polar-phase metastability → polarization switching	soft mode 22.9 + Pca2⊠ ΔE (HfO⊠)	polarization storage
PCM (amorphous↔crystalline resistance)	melt-quench → resistance-state switching	T_m/T_x/ν (GST pseudo-line)	resistance storage (power-law end)
3D NAND CTF (charge trap)	charge dwell in trap layer → escape	E_A/τ/A=1/τ (trap energy spectrum)	charge retention (exponential end)

Unified criterion coordinate (industry material-selection positioning): - ZrO⊠ (soft mode 20.3 + AFE) → DRAM capacitor (charge) - HfO⊠/HZO (soft mode 22.9 + FE) → ferroelectric memory (polarization) - GST pseudo-line (T_m/T_x/ν) → phase-change memory (resistance, power-law end) - SiN/high-k trap layers (E_A spectrum + A=1/τ) → 3D NAND CTF (charge retention, exponential end)

Honest boundary: the unification is conceptual (force-ratio critical / retention language), not formula-level—S-C1/C2 address displacive transitions (soft-mode C44), S-C3 addresses nucleation-growth type (T_m/T_x/ν), S-C4 addresses charge retention (E_A/A=1/τ), with criterion quantities of different families. The bandgap axis uses DFT relative ordering (OptB88vdW systematically underestimates by ~1-2 eV); absolute thresholds require experimental calibration. The retention spectrum is a conceptual extension—phase change vs charge retention are two realizations of the same A=1/τ coordinate, not the same formula.

6. Conclusion

The industrial practice of DRAM high-k dielectrics maps onto a 2D criterion space: soft-mode axis (transition activity) × bandgap axis (leakage wall). Tetragonal ZrO⊠ is the unique Pareto-optimal phase (soft mode 20.3/bandgap 4.03); the 30-year industrial pinning of the tetragonal phase receives a criterion explanation; doping-pinning is quantifiable (C44=83.7·r−30.1, R^2=0.9998); the martensitic mirror explains the dual industry semantics of the criterion. Criterion-D data engineering makes the criterion semantics explicit as "0K instability signal of a real metastable phase," with the data asset (546-compound list) released alongside. This 2D criterion space further extends into the five-dimensional unified storage-medium criterion (transition activity × transition-activity signature × transition limit × retainability/post-drift × retention quality), covering DRAM high-k, ferroelectric memory, phase-change memory, and charge-trap memory. The retention of charge-trap memory (3D NAND CTF) is experimentally confirmed to be a distribution of adsorption strength $A=1/\tau$ (widely distributed time constants), forming the exponential end of a decay-shape dichotomy whose power-law end is the drift of phase-change memory—moving storage-dielectric selection from trial-and-error to criterion-space positioning, with the unified criterion extending from a transition spectrum to a retention spectrum.

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